SMU Lyle School of Engineering

# Reinforcement Learning

# Deep Reinforcement Learning

Based on An Introduction to Deep Reinforcement Learning**
+ Sutton/Barto* Sections 9.7 and 15.7

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*Sutton and Barto, Reinforcement Learning: An Introduction, 2nd edition, MIT Press, Cambridge, MA, 2018

**Vincent François-Lavet, Peter Henderson, Riashat Islam, Marc G. Bellemare and Joelle Pineau (2018), "An Introduction to Deep Reinforcement Learning", *Foundations and Trends in Machine Learning*: Vol. 11, No. 3-4. DOI: 10.1561/2200000071.

Online Material

# Topics of this Course

- Introduction to reinforcement learning
- Markov decision processes
- Part I: Tabular Methods
  - Dynamic programming
  - Monte Carlo methods
  - Temporal-difference learning
  - Multi-step bootstrapping
  - Planning and learning with tabular methods
- Part II: Approximate Solution Methods
  - Prediction and Control using Approximation
  - Eligibility Traces
  - Policy Gradient Methods
- **Part III: Modern RL Methods**
  - **Deep Reinforcement Learning**
  - Current Applications

# Summary of Notation

**General**

| | |
|---|---|
| $X$ | capital letters: random variables |
| $x, p$ | lower-case letters: realizations of random variables or scalar functions |
| $\boldsymbol{w}$ | Bold lower-case letters: real-valued vectors (even if random variables) |
| $\boldsymbol{W}$ | bold capitals: matrices |
| $\alpha$ | Greek letters: parameters (vectors if in bolt) |
| $\Pr\{X = x\}$ | probability that a random variable $X$ takes on the value $x$ |
| $X \sim p$ | random variable $X$ selected from distribution $p(x) = \Pr\{X = x\}$ |
| $\mathbb{E}[X]$ | expectation of a random variable $X$, i.e., $\mathbb{E}[X] = \sum_x p(x)x$ |
| $\text{argmax}_a f(a)$ | a value of action $a$ at which $f(a)$ takes its maximal value |

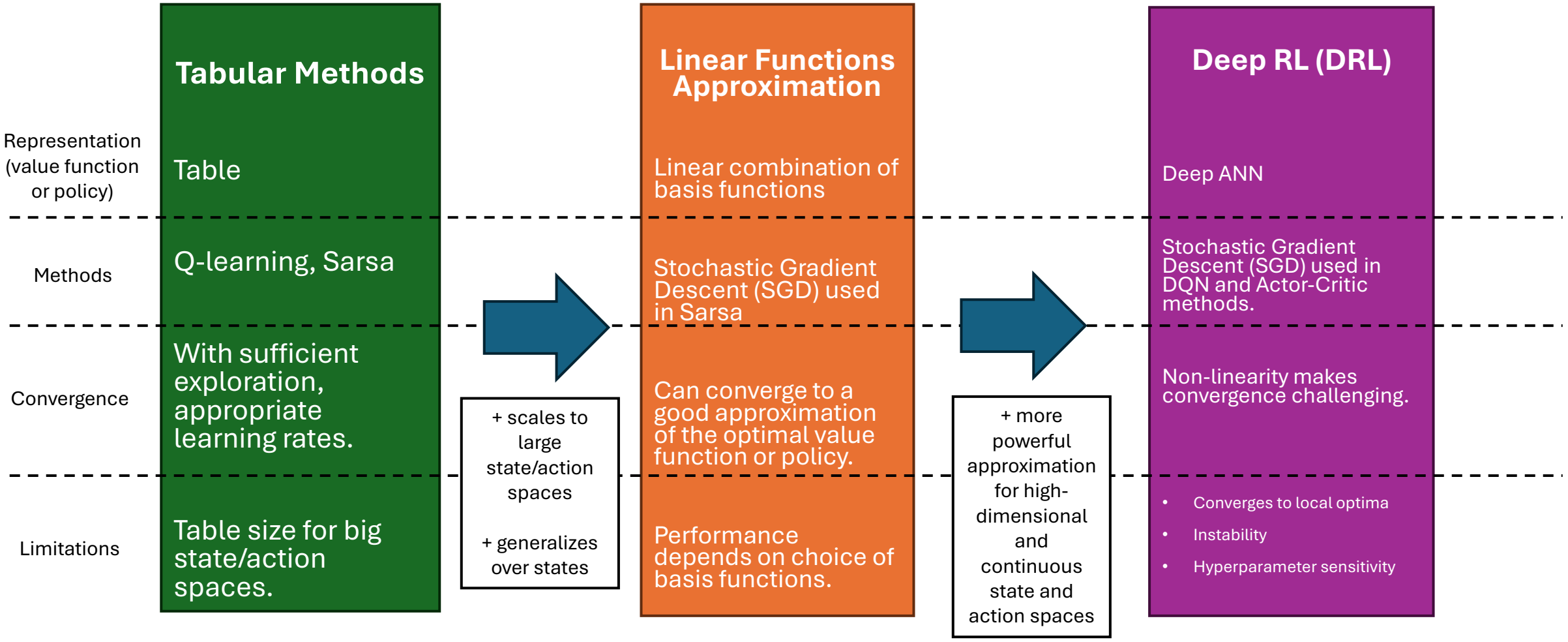
**Value Function**

| | |
|---|---|
| $G_t$ | return (cumulative reward) following time $t$ |
| $G_{t:h}$ | return from $t$ to $h$ (discounted and corrected) |
| $v_\pi(s)$ | value of state $s$ under policy $\pi$ (expected return) |
| $v_*(s)$ | value of state s under the optimal policy |
| $q_\pi(s, a)$ | value of taking action $a$ in state $s$ under policy $\pi$ |
| $q_*(s, a)$ | value of taking action $a$ in state $s$ under the optimal policy |
| $V, V_t$ | array estimates of state-value function $v_\pi$ or $v_*$ |
| $Q, Q_t$ | array estimates of action-value function $q_\pi$ or $v_*$ |

**MDP**

| | |
|---|---|
| $s, s'$ | states |
| $a$ | an action |
| $r$ | a reward |
| $\mathcal{S}$ | set of all (nonterminal) states, $\mathcal{S}^+$ are all states |
| $\mathcal{A}(s)$ | set of all actions available in state $s$ |
| $\gamma$ | discount-rate parameter |
| $t$ | discrete time step |
| $T$ | final time step of an episode (a.k.a. horizon) |
| $A_t$ | random variable for the action at time $t$ |
| $S_t$ | random variable for the state at time $t$ |
| $R_t$ | random variable for the reward at time $t$ |
| $p(s', r \mid s, a)$ | probability of transition to state $s'$ and receiving reward $r$, from state $s$ taking action $a$. |
| $p(s' \mid s, a)$ | probability of transition to state $s'$ fom state $s$ taking action $a$. |
| $r(s, a)$ | expected immediate reward from state $s$ after action $a$. |
| $r(s, a, s')$ | expected immediate reward from state $s$ to $s'$ with action $a$. |
| $\pi(a \mid s)$ | probability of taking action $a$ in state $s$ under stochastic policy $\pi$ |
| $\pi(s)$ | action taken in state $s$ under deterministic policy $\pi$ |

# From Tabular RL to Deep RL



# Value-based DRL Methods

Use ANNs to represent a parameterized value function that is updated via gradient descent.

# Remember: Value Functions

- Immediate reward: $r_t = \mathbb{E}_{a\sim\pi(s_t,\cdot)}\, r(s_t, a, s_{t+1})$

- Value functions:
  - $V^\pi(s) = \mathbb{E}[\sum_{k=0}^{\infty} \gamma^k r_{t+k} \mid S_t = s\,, \pi]$
  - $Q^\pi(s,a) = \mathbb{E}[\sum_{k=0}^{\infty} \gamma^k r_{t+k} \mid S_t = s, A_t = a\,, \pi]$
- Bellman equation:

$$Q^\pi(s,a) = \sum_{s'\in\mathcal{S}} p(s,a,s')\left(r(s,a,s') + \gamma Q^\pi\left(s', a = \pi(s')\right)\right)$$

- Optimality condition:

  $V^*(s) = \max_{\pi\in\Pi} V^\pi(s)$ and $Q^*(s,a) = \max_{\pi\in\Pi} Q^\pi(s,a)$
- Optimal policy: $\pi^*(s) = \operatorname{argmax}_{a\in\mathcal{A}} Q^*(s,a)$

# New: Advantage Function

- The “advantage” of choosing action $a$ over the baseline given by the state value is

$$A^{\pi}(s,a) = Q^{\pi}(s,a) - V^{\pi}(s)$$

- Positive advantage means action $a$ is better than $\pi(s)$
- For the optimal $\pi^*$:

$$A^{\pi^*}(s,a) = \begin{cases} 0 & \text{if } s = s^* \\ < 0 & \text{otherwise.} \end{cases}$$

- $V, Q$ and $A$ can be estimated using Monte Carlo methods (unbiased).
- $V, Q$ and $A$ can be used to calculate a TD-error.

# Remember: Tabular Q-Learning

- An extremely influential method developed by Watkins (1989).
- Tabular update rule only uses Sars:

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha[\underbrace{R_{t+1} + \gamma \max_a Q(S_{t+1}, a)}_{\text{target}} - \underbrace{Q(S_t, A_t)}_{\text{prediction}}]$$

$$\text{TD-error}$$

- $Q$ directly approximates $q_*$ independent of the behavior policy being followed! → Off-policy
- Converges to the optimal value function:
  - State-action pairs are represented discretely (in a table).
  - All action are repeatedly sampled in all states (keeps exploring).

- **Issue**: $Q(s, a)$ can often not be represented in a table due to too many action-value pairs. Use a parameterized approximation $Q(s, a; \boldsymbol{\theta})$

# Fitted Q-Learning

- Use a parameterized approximation $Q(s, a; \boldsymbol{\theta})$
- Initial setting for $\boldsymbol{\theta}_0$ such that all Q-value are close to 0 (helps with learning).
- Target value at iteration $k$:

$$U_k = r + \gamma \max_{a'} Q(s', a'; \boldsymbol{\theta}_k)$$

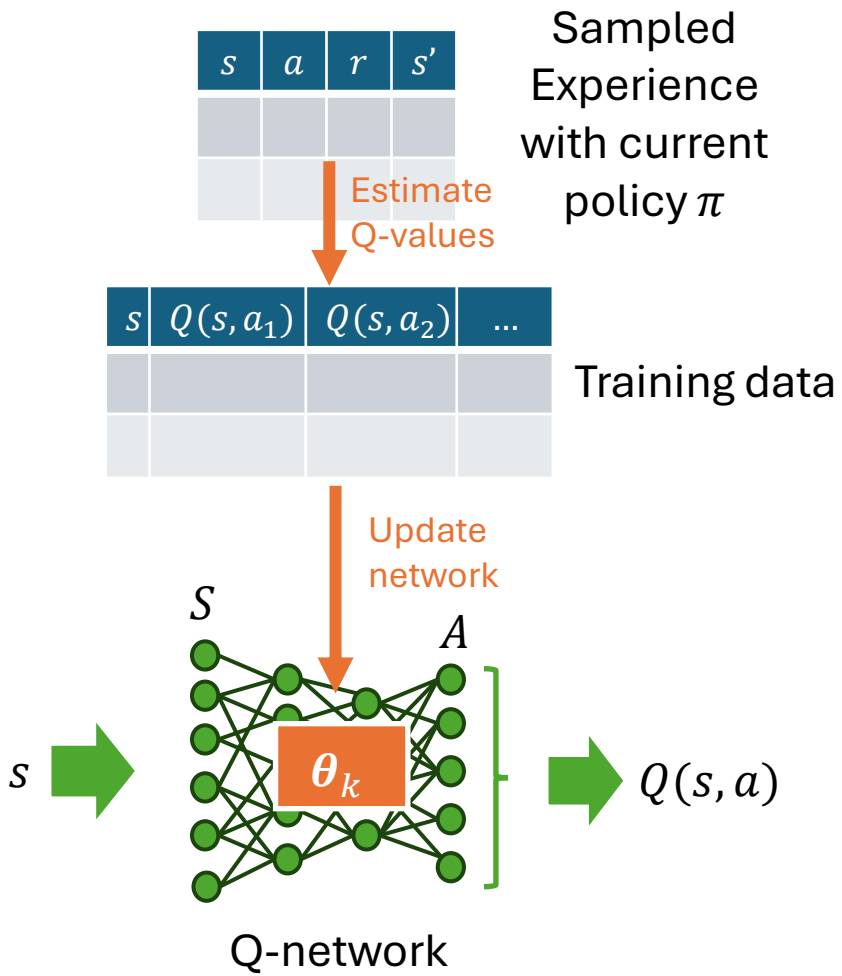
- Update:

$$\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k + \alpha\big(\underbrace{U_k}_{\text{target}} - \underbrace{Q(s, a, \boldsymbol{\theta}_k)}_{\text{prediction}}\big)\nabla Q(s, a, \boldsymbol{\theta}_k)$$

$$\underbrace{\qquad\qquad}_{\text{TD-error}}$$

# Neural fitted Q-learning (NFQ; Riedmiller, 2005)

- Use an ANN called the **Q-network** as the approximator for the Q-function.
- **Issue**: Online learning does not work well.
- **Solution**:
  - Learn the Q-network offline: **batch learning** with experience replay from a sample set of experience containing $(s, a, r, s')$ tuples.
  - Use **supervised learning (regression)** with the sample as the training set.
  - The regression target Q-values are estimated from the tuples using the Bellman equation: $Q(s,a) = r + \gamma \max_{a'} Q(s', a')$
- **More issues**:
  - May not converge or become unstable (uses approximation).
  - Q-values tend to be overestimated due to the max operator in the target.

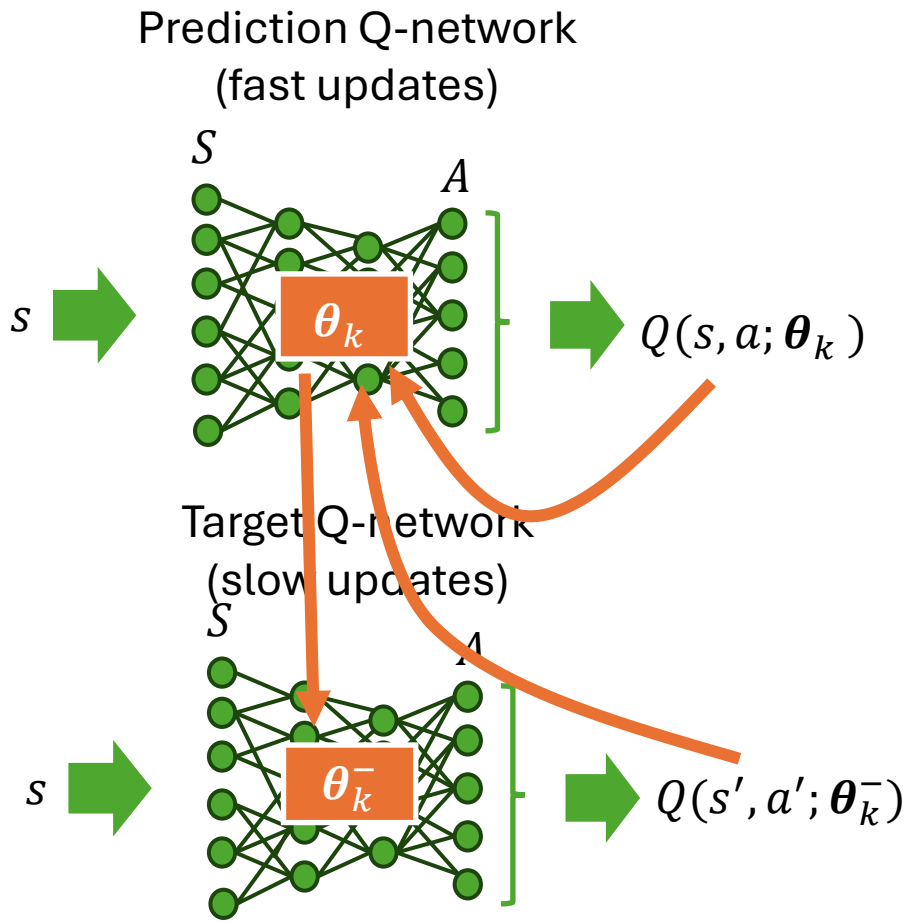


# Deep Q-Network (DQN; Minh et al, 2015)

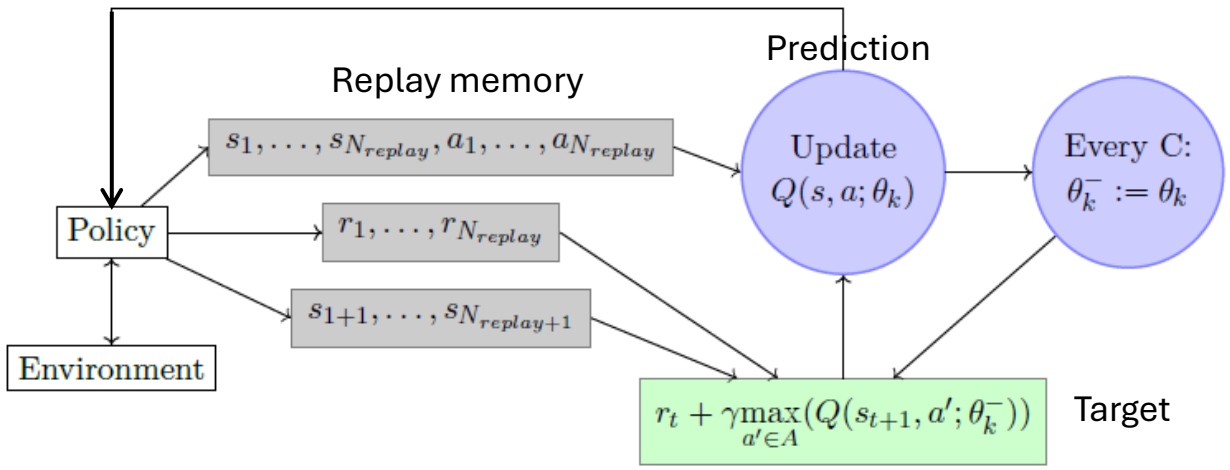
- DNQ addresses the **instability** of NFQ.
- **Idea**: reduce instability by calculating the target from a second Q-network that is updated slowly:

$$U_k = r + \gamma \max_{a'} Q(s', a'; \boldsymbol{\theta}_k^-)$$

- **Updates**:
  - $\boldsymbol{\theta}_k$ is updated every iteration:
    $\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k + \alpha\big(U_k - Q(s, a, \boldsymbol{\theta}_k)\big)\nabla Q(s, a, \boldsymbol{\theta}_k)$
  - $\boldsymbol{\theta}_k^-$ is updated only every $C$ iterations by $\boldsymbol{\theta}_k^- = \boldsymbol{\theta}_k$ .
- **Effect**: Target $U_k$ estimates are kept constant for $C$ iterations, instabilities cannot propagate quickly.



# DQN in Online Settings




1. Initialize $\boldsymbol{\theta}_0$ and $\boldsymbol{\theta}_0^-$ so that all $Q(s, a; \boldsymbol{\theta_0})$ are close to 0.
2. Use an $\epsilon$-greedy policy to collect experience in the form of $N_{replay}$ many $(s, a, r, s')$ tuples.
3. Mini-batches (e.g., 32 elements) are taken randomly from the replay memory to update $\boldsymbol{\theta}_k$.
4. $\boldsymbol{\theta}_k^-$ is only set to $\boldsymbol{\theta}_k$ every $C$ iterations.
5. Go back to collecting more experience with a new policy $\epsilon$-greedy policy extracted from the prediction network.

# Double DQN (DDQN; Van Hasselt, 2016)

- **Issue**: Q-learning and DQN use one function in the target to choose the action and evaluate its value. The max creates an **overestimation bias**.

$$\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k + \alpha \Big( \overbrace{r + \gamma \max_{a'} Q(s', a'; \boldsymbol{\theta}_k)}^{\text{target}} - \overbrace{Q(s, a, \boldsymbol{\theta}_k)}^{\text{prediction}} \Big) \nabla Q(s, a, \boldsymbol{\theta}_k)$$

TD-error

- DDQN **decoupling the action selection and evaluation** by using different parameters:

$$U_k = r + \gamma Q(s', \underset{a}{\operatorname{argmax}}\, Q(s', a'; \boldsymbol{\theta}_k); \boldsymbol{\theta}_k^-)$$

Action evaluation uses more stable $\boldsymbol{\theta}_k^-$

Action selection uses $\boldsymbol{\theta}_k$

# Dueling Network Architecture (Wang, 2015)

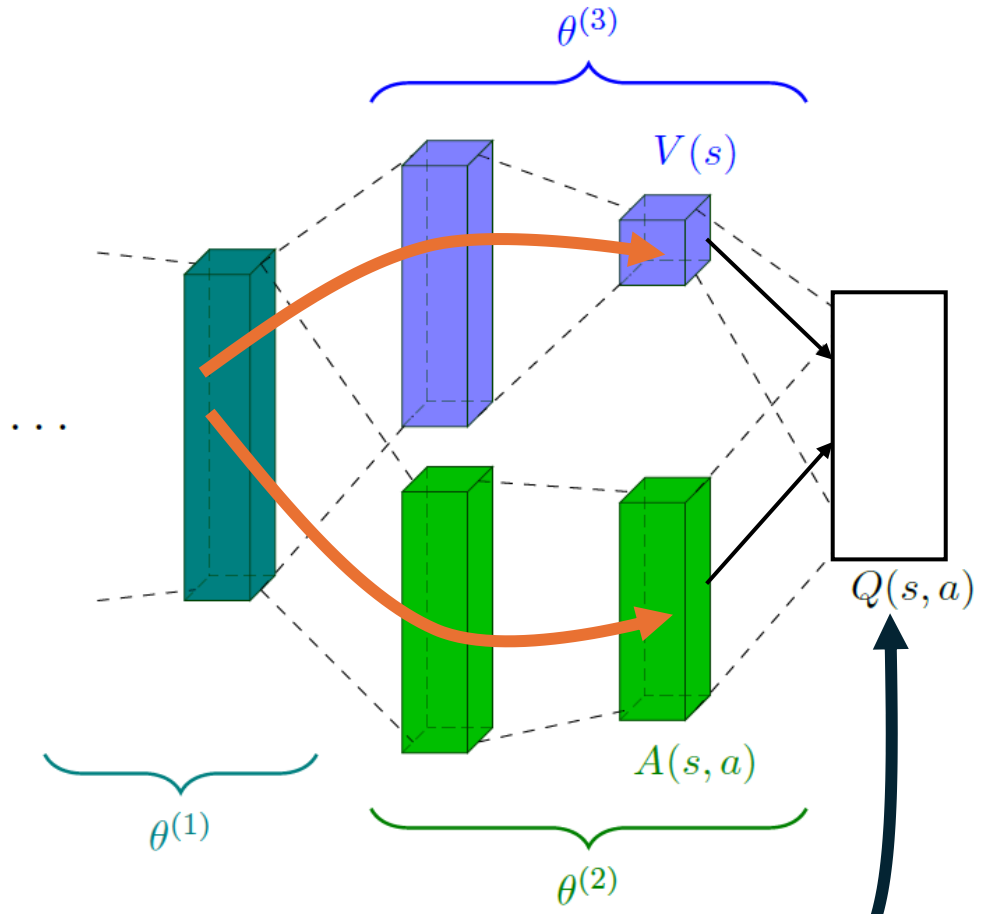
- Dueling Network Architecture only changes the ANN architecture. It learns two components:
  - Value function $V(s; \boldsymbol{\theta}^{(1)}, \boldsymbol{\theta}^{(3)})$
  - Advantage function $A(s, a'; \boldsymbol{\theta}^{(1)}, \boldsymbol{\theta}^{(3)})$
- $Q(s, a)$ is calculated using the equation below.

- Reminder: Advantage function

$$A^{\pi}(s, a) = Q^{\pi}(s, a) - V^{\pi}(s)$$

  gives the "advantage" of choosing action $a$ over the state value.

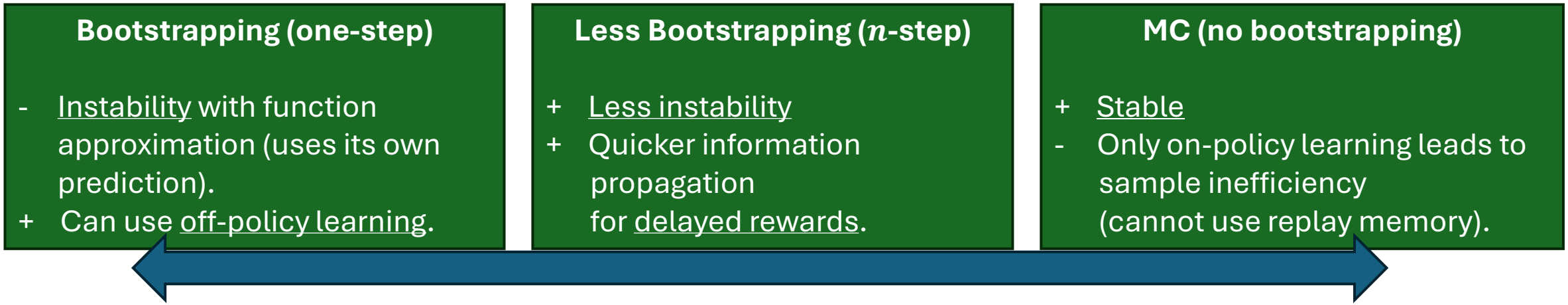
- This approach typically leads to improved performance.




$$Q\left(s, a; \boldsymbol{\theta}^{(1)}, \boldsymbol{\theta}^{(2)}, \boldsymbol{\theta}^{(3)}\right) = V\left(s; \boldsymbol{\theta}^{(1)}, \boldsymbol{\theta}^{(3)}\right) + \left(\mathrm{A}\left(\mathrm{s}, \mathrm{a}; \boldsymbol{\theta}^{(1)}, \boldsymbol{\theta}^{(2)}\right) - \max_{a' \in \mathcal{A}} \mathrm{A}\left(\mathrm{s}, \mathrm{a}'; \boldsymbol{\theta}^{(1)}, \boldsymbol{\theta}^{(2)}\right)\right)$$

Becomes 0 for $a = a^*$

# $n$-step Methods




- ***n*-step** methods are between Q-learning (one-step) and MC, which does not bootstrap.
- Use $n$-step target for DQN-type algorithms:

$$U_k^n = \sum_{t=0}^{n-1} \gamma^t r_t + \gamma^n \max_{a' \in \mathcal{A}} Q(s_n, a'; \boldsymbol{\theta}_k)$$

- **Traces** can also be used.

# Policy Gradient DRL Methods

Use ANNs to represent and learn a parameterized policy.

Note: $\theta$ is called $w$!

# Remember: Policy Gradient Theorem

- We define the performance measure as the true value of following $\pi_\theta$ from the start state $s_0$

$$J(\boldsymbol{\theta}) \stackrel{\text{def}}{=} v_{\pi_\theta}(s_0)$$

- The **policy gradient theorem** gives us an analytical expression for the gradient of the  with respect to the policy parameters:

$$\nabla J(\boldsymbol{\theta}) = \nabla v_{\pi_\theta}(s_0) \propto \sum_s \mu(s) \sum_a q_\pi(s,a) \nabla \pi(a|s,\boldsymbol{\theta})$$

On-policy distribution under $\pi$

Depends on estimates for q

Derived from the def. of the policy.

- The theorem also provides a **strong convergence guarantee**!

Policy: $\pi(a|s,\boldsymbol{\theta}) = \frac{e^{h(s,a,\boldsymbol{\theta})}}{\sum_b e^{h(s,b,\boldsymbol{\theta})}}$

Gradient: $\nabla \ln \pi(s|a,\boldsymbol{\theta}) = h(s,a) - \sum_a' h(s,a')\pi(s|a',\boldsymbol{\theta})$

# Policy Gradient

- Following the ideas of REINFORCE.
- Gradient of a stochastic policy

Needs a value function estimate

$$\nabla_w v_{\pi_w}(s_0) = \mathbb{E}_{s\sim\mu_{\pi_w}, a\sim\pi_w}[\nabla_w (\log \pi_w(s,a)) Q^{\pi_w}(s,a)]$$

Sample state from the on-policy distribution and action from the policy

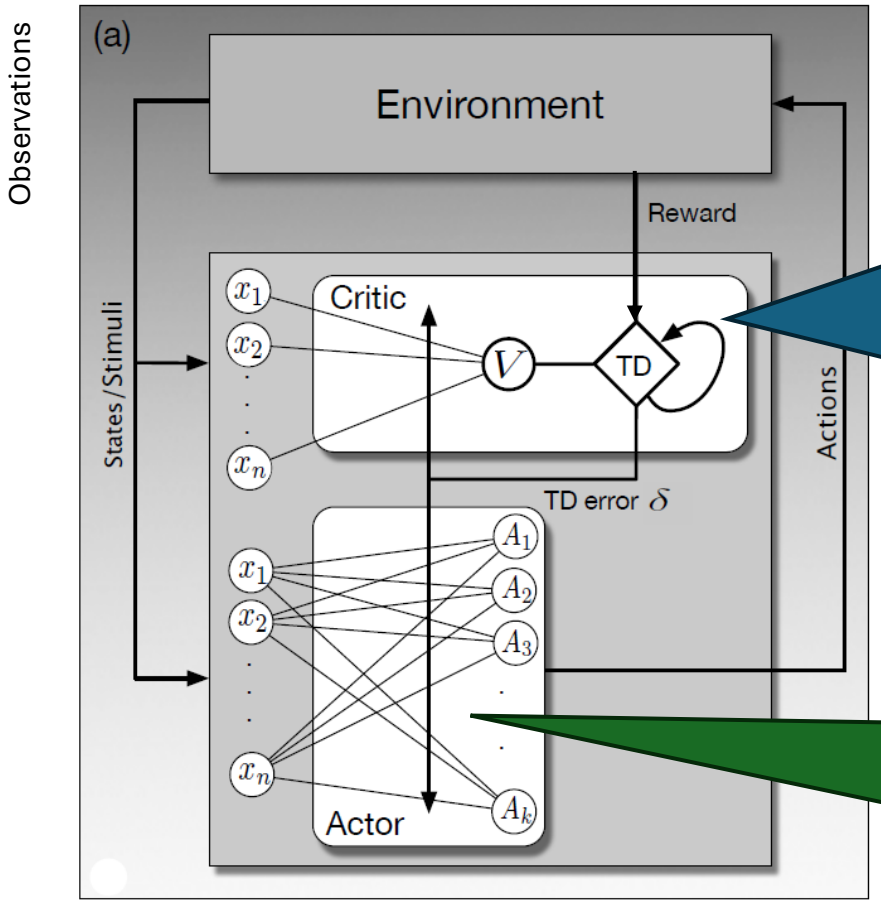
- Gradient of a differentiable deterministic policy

$$\nabla_w v_{\pi_w}(s_0) = \mathbb{E}_{s\sim\mu_{\pi_w}} [\nabla_w(\pi_w) \nabla_a (Q^{\pi_w}(s,a)) |a = \pi_w(s)]$$

Also requires this gradient!

# Neural Actor-Critic Method

- **Remember**: Estimate the value function and a parameterized policy at the same time.




**Critic**

- Represents value function $Q^{\pi_w}(s,a) \approx Q(s,a;\boldsymbol{\theta})$
- Produces the TD error from the reward signal using its network's state-value prediction.
- Adjusts state-value network using the TD error.
- Does not select actions!

**Actor**

- Represents policy $\pi_w$
- Selects actions with a policy network.
- Adjusts a policy network based on the TD error
- No access to reward signal!

Section 15.7 in Sutton and Barto (2018)

# Neural Actor-Critic Method: One-Step Algorithm

**One-step Actor–Critic (episodic), for estimating $\pi_{\boldsymbol{\theta}} \approx \pi_*$**

Input: a differentiable policy parameterization $\pi(a|s,\boldsymbol{\theta})$ — actor

Input: a differentiable state-value function parameterization $\hat{v}(s,\mathbf{w})$ — critic

Parameters: step sizes $\alpha^{\boldsymbol{\theta}} > 0$, $\alpha^{\mathbf{w}} > 0$

Initialize policy parameter $\boldsymbol{\theta} \in \mathbb{R}^{d'}$ and state-value weights $\mathbf{w} \in \mathbb{R}^{d}$ (e.g., to $\mathbf{0}$)

Loop forever (for each episode):

  Initialize $S$ (first state of episode)

  $I \leftarrow 1$ — Takes care of discounting $\gamma^t$

  Loop while $S$ is not terminal (for each time step):

    $A \sim \pi(\cdot|S,\boldsymbol{\theta})$

    Take action $A$, observe $S', R$

    $\delta \leftarrow R + \gamma \hat{v}(S',\mathbf{w}) - \hat{v}(S,\mathbf{w})$ — TD error: how much is $G$ better than $\hat{v}$

    (if $S'$ is terminal, then $\hat{v}(S',\mathbf{w}) \doteq 0$)

    $\mathbf{w} \leftarrow \mathbf{w} + \alpha^{\mathbf{w}} \delta \nabla \hat{v}(S,\mathbf{w})$ — Update the critic (value function) = $TD(0)$

    $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \alpha^{\boldsymbol{\theta}} I \delta \nabla \ln \pi(A|S,\boldsymbol{\theta})$ — Update the actor (policy) = REINFORCE

    $I \leftarrow \gamma I$

    $S \leftarrow S'$

Possible improvements:

- Sample efficiency: Use a replay memory with a set of $(s, a, r, s')$ tuples.
- Stability and faster reward propagation: Multi-step methods

# Natural Policy Gradients (NPG)

- Regular policy gradient methods update policy parameters in the direction of the steepest ascent of the performance measure $J(\boldsymbol{w}) \stackrel{\text{def}}{=} v_{\pi_w}(s_0)$ given a small change of the parameter vector $\boldsymbol{w}$.
- This is achieved by an update $\Delta \boldsymbol{w} \propto \nabla_w J(\boldsymbol{w})$ which maximizes $J(\boldsymbol{w}) - J(\boldsymbol{w} + \Delta \boldsymbol{w})$ under a constraint on $\|\Delta \boldsymbol{w}\|_2$
- **Issue**: The same amount of change in the parameter space $\boldsymbol{w}$ can lead to very different magnitudes of change in the actual policy $\pi_w$ (the probability distribution over actions) resulting in instability.

- **Idea**: Natural gradients define the steepest direction by the largest change in the value function given a small change in the probability distribution of the policy $\pi_w$.
- This requires redefining the constraint on $\Delta \boldsymbol{w}$. Kullback-Leibler (KL) divergence measures the difference between two policies' action probability distributions $D_{KL}(\pi_w || \pi_{w+\Delta w})$.
- A local approximation, the Fisher information criterion, can be used to adjust the gradient for policy changes:

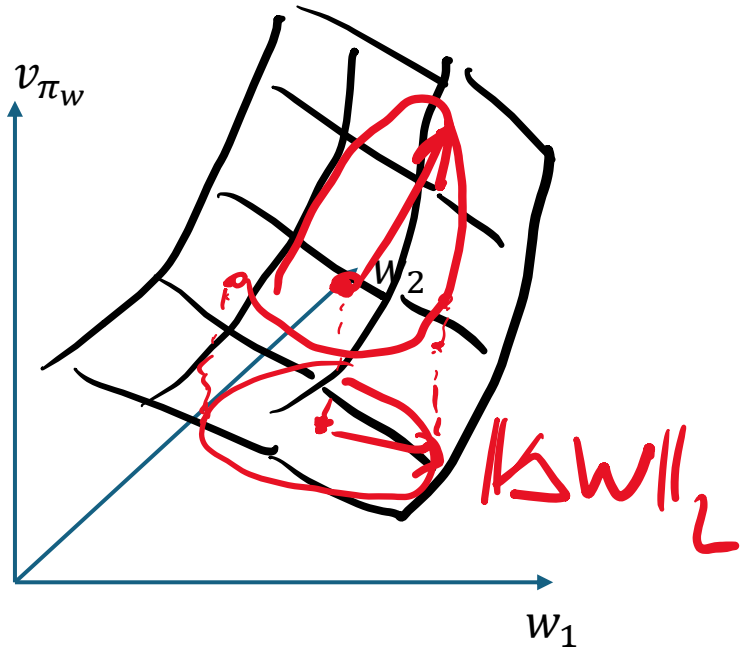
$$\Delta \boldsymbol{w} \propto F_w^{-1} \nabla_w J(\boldsymbol{w})$$

- $F_w$ is the Fisher information matrix that depends

- **Problem**: Calculating the Fisher information matrix of deep learning models is too expensive.
- The idea inspired several state-of-the-art methods:
  - Trust Region Optimization (TRPO)
  - Proximal Policy optimization (PPO)

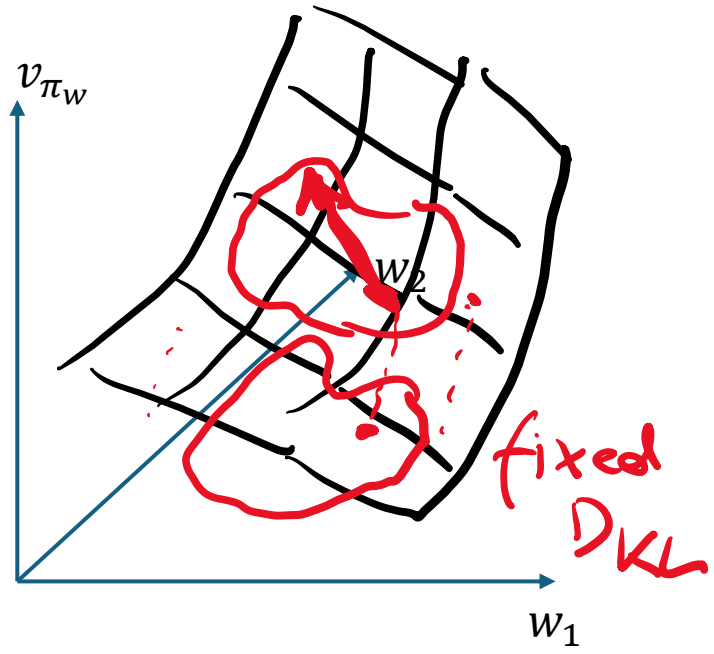
# Standard vs. Natural Policy Gradient

Standard Gradient

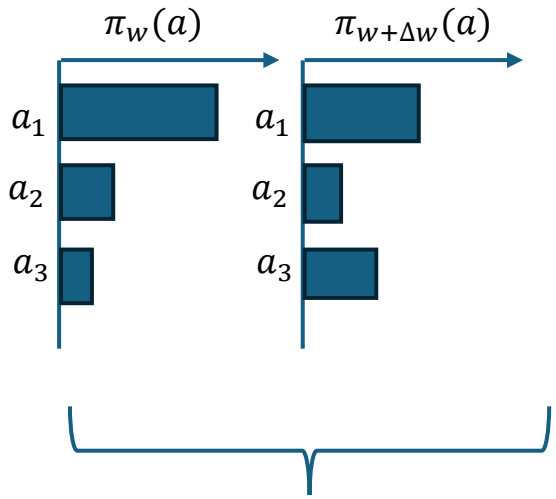



Move a fixed distance in the **parameter space**. This can lead to large policy changes.

Natural Policy Gradient




Move a fixed distance in the **policy space**.




$D_{KL}(\pi_w||\pi_{w+\Delta w})$

# Trust Region Policy Optimization (TRPO; Schulman et al, 2015)

- Simplification of NPG.
- Finds the update that improves the advantage function the most

$$\max_{\Delta w} \mathbb{E}_{s\sim\mu^{\pi_w}, a\sim\pi_w}\left[\frac{\pi_{w+\Delta w}(s,a)}{\pi_w(s,a)} A^{\pi_w}(s,a)\right]$$

Prefer new policies that pick high-advantage actions.

- constrained by an explicitly bound the policy change per update step.

$$\mathbb{E}\left[D_{KL}(\pi_w(s,\cdot)||\pi_{w+\Delta w}(s,\cdot))\right] \leq \delta \quad \text{where } \delta \in \mathbb{R}$$

- $\delta$ is a tuneable hyperparameter specifying the size of the "trust region."

# Proximal Policy Optimization (PPO; Schulman et al, 2017)

- A popular and more practical variant of TRPO.
- Instead of KL-divergence, it uses a clipped surrogate objective function to penalize overly large policy changes with at probability ratio outside of $[1-\epsilon, 1+\epsilon]$

$$r_t(w) = \frac{\pi_{w+\Delta w}(s,a)}{\pi_w(s,a)}$$

$$\max_{\Delta w} \mathbb{E}_{s\sim\mu^{\pi_w}, a\sim\pi_w}\left[\min\left(r_t(w)A^{\pi_w}(s,a), \text{clip}(r_t(w), 1-\epsilon, 1+\epsilon)A^{\pi_w}(s,a)\right)\right]$$

- $\epsilon$ is a tunable hyperparameter
- Achieves much of the stability of NPG methods with the computational efficiency of first-order optimization.

# Generalization

# Generalization in RL

a) Perform well in an environment after limited data is gathered
   (= sample efficiency)

b) Perform well in a new, but related environment
   (~ transfer learning)